30. Stratified Sampling

March 27, 2026 9:46 AM

Generally: Stratified Sampling in Statistics means breaking up your population / Domain into strata (pieces) and sampling for each of them.

* Age Strata: • $n = 100$ → sample all people at random

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• $n_1 = \dots = n_5 = 20$ sample each age range at random: $[0,15)$, $[15,25)$, $[25,40)$, $[40,60)$, $[60,\infty)$

In Stat 413, we will break our Domain of Integration into pieces / Strata and do Monte Carlo integration on each piece.

Why do we do this? ⇒ Decrease the Variance of our estimate.

Mathematically, partition $[0,1]$ into K disjoint subsets $A_1, \dots, A_K$ s.t. $\bigcup_{i=1}^{K} A_i = [0,1]$

$$\mu = \int_0^1 g(x)\,dx = \sum_{i=1}^{K} \int_{A_i} g(x)\,dx \quad \leftarrow \text{Calc world}$$

probability world

$$= \sum_{i=1}^{K} \omega_i \int_{A_i} g(x) \frac{dx}{\omega_i} = \boxed{\sum_{i=1}^{K} \omega_i \, \mathbb{E}_{A_i}[g(X)]}$$

where $\omega_i = |A_i| = \int_{A_i} dx$, $\sum \omega_i = 1$

weighted sum of Expected Values

The point of writing $\mathbb{E}_{A_i}[g(X)]$ is that we can estimate each of these "means" via a sample average / LLN.

Estimator: if we were to sample n_i points uniformly from each A_i, x_{ij}, $j = 1, \dots, n_i$

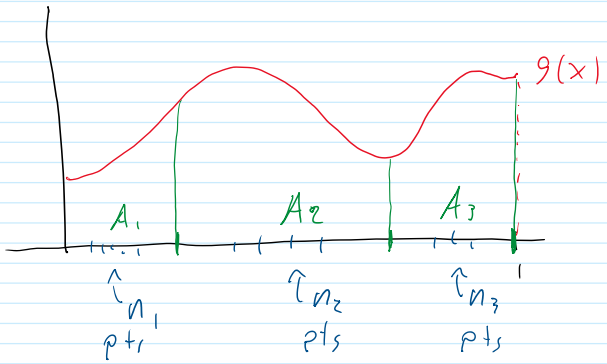
then $$\hat{\mu}_n = \boxed{\sum_{i=1}^{K} \omega_i \times \frac{1}{n_i} \sum_{j=1}^{n_i} g(x_{ij})}$$

Then $\mu_n = \boxed{\sum_{i=1} \; \sum_{j=1} g(x_{ij})}$

and $Var(\hat{\mu}_n) = \sum_{i=1}^{k} \frac{\omega_i^2}{n_i} Var_{A_i}(g(x))$

Variance of the usual MC estimator.

Question: is this less than $\frac{1}{n} Var(g(x))$



Total Sample size

$$n = \sum_{i=1}^{k} n_i$$

Example: $g(x) = 2$

How many points do I need to sample to get a good estimate of $\int_0^1 g(x)\,dx$?

$$2 = \int_0^1 g(x)\,dx = \frac{1}{n}\sum_{i=1}^{n} g(x_i) = \frac{1}{n}\sum_{i=1}^{n} 2 = 2, \; \forall n$$

More Interesting Example:

Intuition:

choose $n_1 \ll n_2$

(we usually treat $n = n_1 + n_2$ as fixed)

Point: if $g(x)$ is highly varying in some region then sample a lot!

if $g(x)$ is close to constant, don't sample a lot.

Lets say that we sample $n_i = \omega_i n$, which is that we choose n_i proportional to $\omega_i = |A_i|$

assume integer

Claim: Variance will set smaller under this sampling

say $\hat{\mu}_n^{SS}$ = Stratified est.

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$\hat{\mu}_n$ = MC est.

$$Var(\hat{\mu}_n^{SS}) = \sum_{i=1}^{k} \frac{\omega_i}{n_i} Var_{A_i}(g(x))$$

$$= \sum_{i=1}^{k} \frac{\omega_i}{\omega_i n} Var_{A_i}(g(x))$$

$$= \frac{1}{n} \sum_{i=1}^{k} \omega_i \mathbb{E}_{A_i}\left[\left(g(x) - \mathbb{E}_{A_i}[g(x)] \right)^2 \right]$$

$$- \mathbb{E}\, g(x) + \mathbb{E}\, g(x)$$

$$= \frac{1}{n} \sum_{i=1}^{k} \omega_i \left[\underbrace{\mathbb{E}_{A_i}\left[(g(x) - \mathbb{E}\, g(x))^2 \right]}_{Var(\hat{\mu}_n)} - \left(\mathbb{E}_{A_i}[g(x)] - \mathbb{E}\, g(x) \right)^2 \right]$$

$$= Var(\hat{\mu}_n) - \underbrace{\left(\mathbb{E}_{A_i}[g(x)] - \mathbb{E}\, g(x) \right)^2}_{Var(\hat{\mu}_n^{SS}) \le Var(\hat{\mu}_n)}$$

*Note: the rate of convergence is still $\sim O(n^{-1/2})$